```
Function f(x)
    // Define the function to be integrated
    // Replace this with the actual function

Function simpsons_3_8_rule(a, b, n)
    h = (b - a) / n
    sum = f(a) + f(b)
    for i = 1 to n-1
        if i % 3 == 0
            sum = sum + 2 * f(a + i * h)
        else
            sum = sum + 3 * f(a + i * h)
        result = (3 * h / 8) * sum
    return result

Function main()
    // Input the interval [a, b] and the number of subintervals (n)
    a = input("Enter the lower limit of the interval (a): ")
    b = input("Enter the upper limit of the interval (b): ")
    n = input("Enter the number of subintervals (must be a multiple of 3): ")

    // Check if n is a multiple of 3
    if n % 3 != 0
```

```
    print("Error: Number of subintervals must be a multiple of 3.")
    return

// Calculate and output the result
result = simpsons_3_8_rule(a, b, n)
print("The result of the integration using Simpson's 3/8th rule is: ", result)
```